

Statistics

Def:- Statistics is the science of collecting, organizing and analyzing data.

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Decision making

Data $\Rightarrow$ Piece of Information

Eg:- Weights of students in a class
60 Kg, 50 Kg, 45 Kg, 30 Kg...

Data Analyst Avg weight 70-80 Kg
Data Scientist

$\rightarrow$ Previous Data

House Price Prediction

I/p $\rightarrow$ O/p

City	Area (sqft)	No. of Rooms	O/p
BLR	2500	3	45 Laks
HYD	1500	2	50 Laks
Chennai	500	4	—

Importance of stats

- i) Data Exploration $\Rightarrow$ Summarize
- ii) Model Building $\Rightarrow$ Validation
- iii) Statistical Analysis $\Rightarrow$ Sample Data
Population Data
- iv) Hypothesis Testing
- v) Optimization and Efficiency